

Peer's Charity Academic Study Guides

C++ Pointers & Memory Management Cheat Sheet | CSE 2105 High-Yield Review

1. Pointer Basics & Arithmetic

- `int* p = &var;` (Stores address of var)
- `*p = 50;` (Dereferences pointer to update value)
- Pointer arithmetic: `p + 1` increments address by `sizeof(T)` bytes.

2. Dynamic Memory Management (C++ vs C)

- Allocation: `int* arr = new int[100];` (Calls constructors, type-safe)
- Deallocation: `delete[] arr;` (Must use array delete to prevent leaks!)

3. Modern Smart Pointers (`std::memory`)

- `std::unique_ptr<T>`: Exclusive ownership. Non-copyable, movable. Auto freed when out of scope.
- `std::shared_ptr<T>`: Shared ownership using reference count control block.
- `std::weak_ptr<T>`: Non-owning observer, prevents cyclic references and deadlocks.

4. Best Practices for Exams

- Always initialize pointers (`nullptr`). Never leave raw pointers uninitialized.
- Use RAII: Wrap resource lifetime in constructor/destructor wrappers.

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